

Mid-Semester Progress Report

Green Fuel Intelligence System (GFIS): A Physics-Guided AI Digital Twin for Methane Yield Prediction and Stability Monitoring in Anaerobic Digestion

Course	DISSERTATION (Tentative NSP 2)
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Abstract

This mid-semester report presents the progress of the Green Fuel Intelligence System (GFIS), a physics-guided AI digital twin for methane-yield prediction and stability monitoring in anaerobic digestion. The abstract presentation was accepted by faculty with an Excellent remark. The evaluator specifically recognized the work as moving beyond black-box machine learning toward physics-informed machine learning, soft sensing, and digital-twin simulation, and the evaluator's PIML framing has been adopted as a key positioning guide for the mid-semester prototype. Here, PIML is used in its established meaning of Physics-Informed Machine Learning while acknowledging that the evaluator's feedback sharpened the terminology and research direction. In response, the current implementation has been organized as a working research prototype with a data pipeline, XGBoost tabular model, PyTorch LSTM branch, ensemble prediction, volatile-solids-based physics feasibility check, VFA/ALK soft sensor, FastAPI backend, Streamlit dashboard, browser-based control room, industrial simulation page, SQLite logging, Docker/AWS deployment preparation, and ROS2/Gazebo extension planning. The current evaluation on the synthetic anaerobic digestion dataset gives a tabular model R^2 of 0.7556, RMSE of 6.7933, and MAE of 5.5517, with zero physics-bound violations in the physics-guided ensemble. Controlled 48-hour scenario experiments demonstrate nominal operation, organic overload warning, pH shock, low-temperature disturbance, reduced HRT washout risk, high-solids feedstock, and corrective recovery operation. The remaining work will focus on strengthening the digital twin with higher-quality data, uncertainty estimation, additional benchmark models, richer explainability, and final dissertation-grade validation.

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1. Background and Accepted Abstract Status

Anaerobic digestion converts organic waste into methane-rich biogas, but methane yield depends on interacting biological, chemical, and operational variables such as temperature, pH, organic loading rate, hydraulic retention time, total solids, volatile solids, carbon-to-nitrogen ratio, moisture, and prior reactor state. GFIS is designed to support this process through predictive modeling, process stability estimation, and digital-twin simulation rather than by returning only a single black-box prediction.

The abstract/outline presentation was accepted in the BITS abstract evaluation stage. The evaluator marked the work as Excellent and appreciated the shift from black-box machine learning to physics-informed machine learning. The evaluator also emphasized the PIML framing, which this report uses as a project-positioning term for Physics-Informed Machine Learning rather than as a claim of inventing the acronym. The same feedback suggested using high-quality industrial data, responding to current discussion on digital twins, and strengthening the project as a simulation dashboard rather than a simple web prediction tool. This mid-semester report is therefore positioned as the technical response to that feedback.

Evaluator point	GFIS response	Mid-sem evidence
Move beyond black-box ML	Use physics-guided feasibility checking and explicit violation reporting.	Physics violation count is zero in current evaluation.
PIML terminology and research framing	Adopt the evaluator's PIML framing as the mid-sem positioning for Physics-Informed Machine Learning.	Report, portal, and simulator now present GFIS as a PIML-guided digital-twin prototype.
Predict hard-to-measure chemical state	Implement VFA/ALK soft sensor and stability label output.	Stable/Warning/Critical outputs are generated by API and dashboards.
Use real/high-quality industrial data	Keep current synthetic data as controlled prototype and plan real data ingestion pipeline.	Remaining work includes public/lab/IoT/SCADA data calibration.
Build digital twin, not only web prediction	Expose plant-run traces, what-if simulation, optimization, and industrial simulator.	48-hour scenario experiments and local demo surfaces are live.

2. Work Completed up to Mid-Semester

- Created a structured GFIS codebase with ``gfis/``, ``backend/``, ``dashboard/``, ``frontend/``, ``data/``, ``models/``, ``reports/``, ``deployment/``, and ``tests/`` folders.
- Generated synthetic anaerobic digestion dataset with process variables, methane yield, VFA/ALK ratio, and stability labels.
- Implemented preprocessing with missing-value handling, scaling, lag features, rolling means, process ratios, and interaction variables.
- Implemented XGBoost tabular methane-yield prediction with fallback model support.
- Implemented PyTorch LSTM training pipeline for future time-series plant data.
- Implemented ensemble prediction and physics-guided feasible-bound checking.
- Implemented VFA/ALK soft sensor and stability classification.
- Exposed FastAPI endpoints for prediction, simulation, plant run, soft sensor, optimization, evaluation, stress tests, and training.
- Built Streamlit dashboard, HTML model control room, and industrial simulation page.
- Generated evaluation report, demo evidence, and controlled scenario experiments.
- Prepared Docker, AWS App Runner, Ubuntu VM, and ROS2/Gazebo extension notes.

3. Current System Architecture

GFIS is organized as a layered digital-twin system. The current architecture deliberately separates data, models, physics rules, soft sensing, simulation, interfaces, and deployment so that future plant data can be attached without rewriting the full system.

Layer	Implemented function	Status
Data layer	Synthetic AD data, raw/processed CSV, SQLite logging	Working
Preprocessing layer	Scaling, lag features, rolling means, ratios, interactions	Working
AI layer	XGBoost tabular model, PyTorch LSTM branch, ensemble	Working
Physics layer	VS-based methane upper-bound and violation reporting	Working
Soft sensor layer	VFA/ALK estimation and stability classification	Working
Digital twin layer	What-if simulation, 48-hour plant run, optimization	Working
Interface layer	FastAPI, Streamlit, browser control room, industrial simulator	Working
Deployment layer	Docker, AWS App Runner notes, Ubuntu/ROS2/Gazebo plan	In progress

4. Model Evaluation and Research Evidence

The current model evaluation uses a synthetic anaerobic digestion dataset with 2400 samples and 24 training epochs for the LSTM branch. The tabular model currently performs strongest, while the LSTM branch is retained for future time-series plant data where temporal retention effects are expected to become more important.

Model	R2	RMSE	MAE
LSTM	0.2942	11.5712	9.0685
XGBRegressor	0.7556	6.7933	5.5517
Validation-weighted ensemble	0.7554	6.8114	5.5699
Physics-guided ensemble	0.7554	6.8114	5.5699

Physics violation count during evaluation: 0. This is important because the physics-guided layer does not simply improve numerical metrics; it audits whether predictions remain feasible under the volatile-solids methane bound.

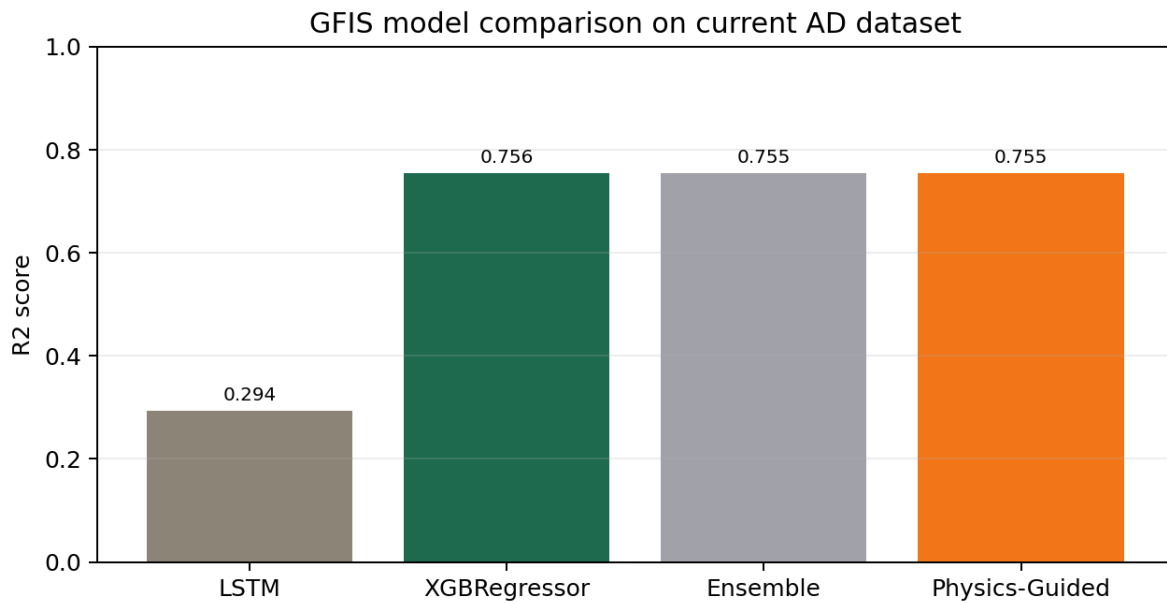


Figure 1. Current GFIS model comparison.

5. Digital Twin Scenario Experiments

The digital-twin experiments evaluate plant behavior over controlled 48-hour windows. These scenarios demonstrate that GFIS can support operating decision analysis rather than only return point predictions.

Scenario	Mean methane	Max VFA/ALK	Final stability	Warning/Critical hours
Nominal mesophilic operation	199.342	0.197	Stable	0
Organic overload	200.932	0.335	Warning	48
pH acidification shock	152.935	0.284	Stable	0
Low-temperature disturbance	183.397	0.227	Stable	0
Reduced HRT washout risk	194.579	0.278	Stable	0
High-solids feedstock	213.273	0.251	Stable	0
Corrective recovery operation	196.719	0.152	Stable	0

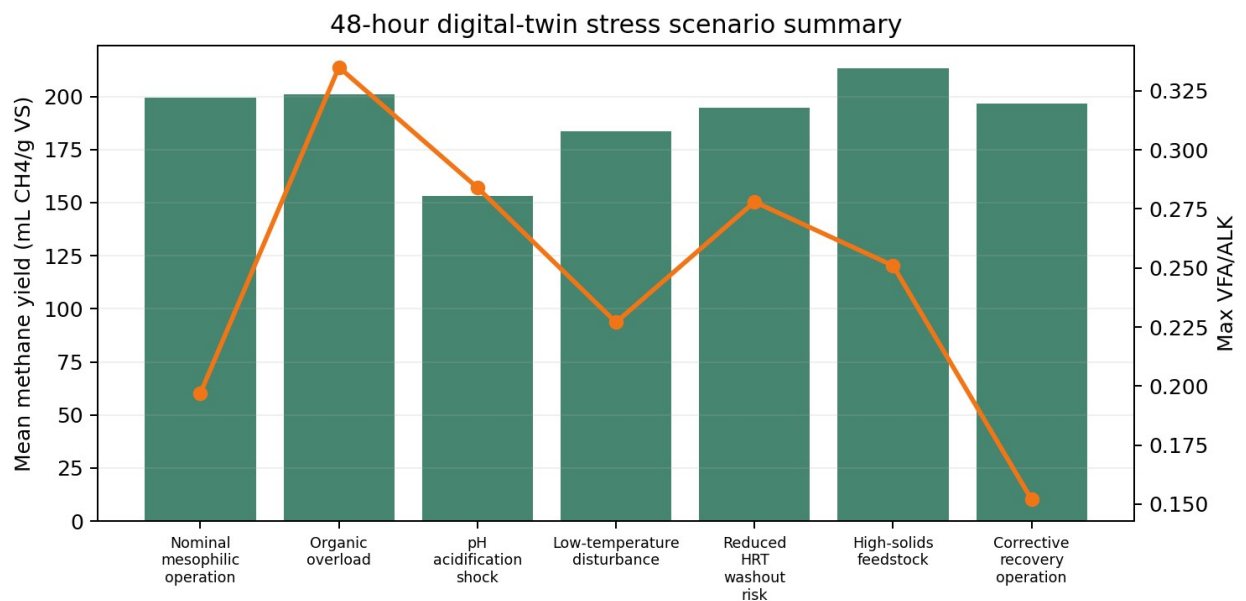


Figure 2. Scenario-level methane and VFA/ALK response across 48-hour plant runs.

6. Local Demonstration and Working Interfaces

The mid-semester prototype is demonstrable locally using the saved model bundle. The system exposes a backend API, a browser-based model control room, an industrial simulation interface, and a Streamlit research dashboard.

Surface	Local URL	Purpose
FastAPI backend	http://127.0.0.1:8010	Core prediction and simulation API
API documentation	http://127.0.0.1:8010/docs	Interactive endpoint testing
Model control room	http://127.0.0.1:8520	Browser digital twin interaction
Industrial simulation	http://127.0.0.1:8520/industrial_simulation.html	48-hour process and plant-level simulation
Streamlit dashboard	http://127.0.0.1:8510	Research dashboard for dataset, model, and scenario review

7. Limitations and Risk Controls

The current prototype is intentionally transparent about its limitations. The data is synthetic/literature-inspired and is suitable for controlled research demonstration, but final claims must not be presented as industrial plant validation until calibrated against public, laboratory, IoT, SCADA, or plant-specific data.

- Synthetic dataset limits external validity; mitigation is to add real or public AD datasets where available.
- LSTM performance is currently weaker than the tabular model; mitigation is to retain LSTM for genuine time-series data and expand benchmark comparison.
- Physics constraint is an upper-bound feasibility layer; mitigation is to add richer biochemical constraints and literature-backed equations.
- Soft sensor estimates VFA/ALK indirectly; mitigation is to calibrate against measured VFA/ALK values when real data is available.
- Dashboard and API are working prototypes; mitigation is to harden deployment with Docker, database migration, authentication, and logging.

8. Remaining Work Plan

Phase	Work item	Expected output	Priority
Mid-sem + 2 weeks	Add benchmark regressors and uncertainty estimate	Expanded metrics table	High
Mid-sem + 3 weeks	Strengthen physics equations and citation support	Final methodology section	High
Mid-sem + 4 weeks	Add or calibrate with public/industrial AD data if available	Dataset comparison evidence	High
Mid-sem + 5 weeks	Capture dashboard/control-room screenshots	Final report figures	Medium
Mid-sem + 6 weeks	Improve SHAP/explainability plots	Feature interpretation section	Medium
Final phase	Package report, viva deck, Turnitin-ready PDF, demo checklist	Final dissertation submission pack	High

9. Conclusion

At mid-semester, GFIS has progressed from an accepted abstract into a working research prototype. The system already demonstrates methane-yield prediction, physics-guided feasibility checking, VFA/ALK soft sensing, stability classification, simulation, optimization, and multiple interfaces. The evaluator's recommendation to position the work as a PIML-guided digital twin has been incorporated directly into the prototype, portal language, and remaining work plan. The most important remaining task is to strengthen validation with better data and richer literature-supported biochemical constraints before final submission.

Appendix A. Evidence and Artifact Index

Evidence	Location
Accepted abstract evaluation screenshot	Phase_Workspaces/01_Abstract_Outline_Submission/Abstract_Viva_Evaluation
Viva transcript/supporting discussion PDF	Phase_Workspaces/01_Abstract_Outline_Submission/Abstract_Viva_Evaluation
Evaluation report	GFIS_Project/reports/evaluation_report.md
Demo evidence	GFIS_Project/reports/demo_evidence.md
Controlled scenario experiments	GFIS_Project/reports/research_scenario_experiments.md
Research report draft	GFIS_Project/reports/GFIS_Dissertation_Research_Report.docx
Deployment notes	GFIS_Project/docs/AWS_DEPLOYMENT_NOTES.md
ROS2/Gazebo extension blueprint	GFIS_Project/docs/ros2_gazebo/integration_blueprint.md

Supervisor Review and Signatures

This section is reserved for review, comments, and signatures if required during mid-semester evaluation.

Signature of Student: _____

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